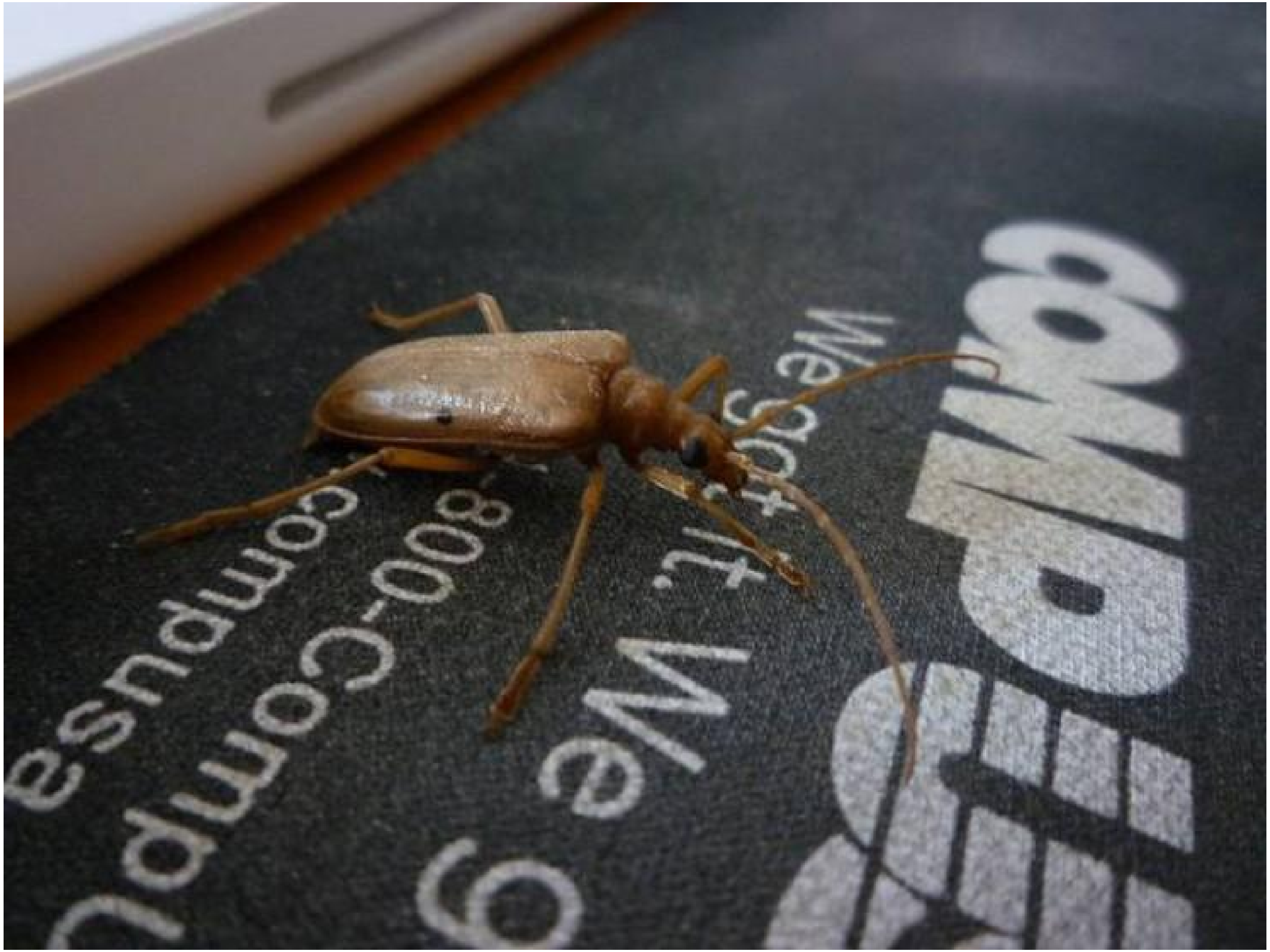
CSE 4513

Lec – 10

Debugging and Unit Testing



What is a Computer Bug

- ✓ a bug refers to an error, fault or flaw in any computer program or a hardware system.
- ✓ produces unexpected results or causes a system to behave unexpectedly.
- ✓ In short, it is any behavior or result that a program or system gets but it was not designed to do.
- ✓ From a developer perspective, bugs can be syntax or logic errors within the source code of a program.
- ✓ These errors can often be fixed using a development tool suitably named a debugger.

FIRST "COMPUTER BUG"

- ✓ the very first instance of a computer bug was recorded at 3:45 pm (15:45) on the 9th of September 1947.
- ✓ It was a **moth**, and it's been preserved behind some adhesive tape for nearly 70 years.
- ✓ U.S. Navy officer, **Grace Hopper**, a member of Harvard's programming team working on the **Mark II Aiken Relay Calculator**, was working late at night with the windows opened (no screens). The moth—roughly four inches in wingspan—was attracted by the light of the room and heat of the calculator and settled its way into the machine…where it was beaten to death by one of the relays.
- ✓ Hopper's team took the bug out of the relay, taped it to their logbook and labeled it as the "first actual case of a bug being found."


9/9
0800 Anctan started
1000 stopped - anctan ✓ 1.2700 9.037 847 025
9.037 846 995 conect
13" uc (032) MP - MC 2.130476415 4.615925059(-2)
(033) PRO 2 2.130476415
conect 2.130676415
Relays 6-2 in 033 failed special speed test
In relay 11.000 test.
Relays changed
1100 Started Cosine Tape (Sine check)
1525 Started Mult+ Adder Test.
1545 Relay #70 Panel F
(moth) in relay.
First actual case of bug being found.
1630 antangent started.
1700 closed down.


Grace

The very first recorded computer bug

MOST OF THE ART OF PROGRAMMING IS DEBUGGING

- ✓ Bugs are always present in large applications.

- ✓ Most of a software life cycle is in the support phase, so debugging saves big money!

What is debugging?

Things to think about:
- What caused the bug?
- How did you end up finding it?
- How could you have avoided the bug in the first-place?

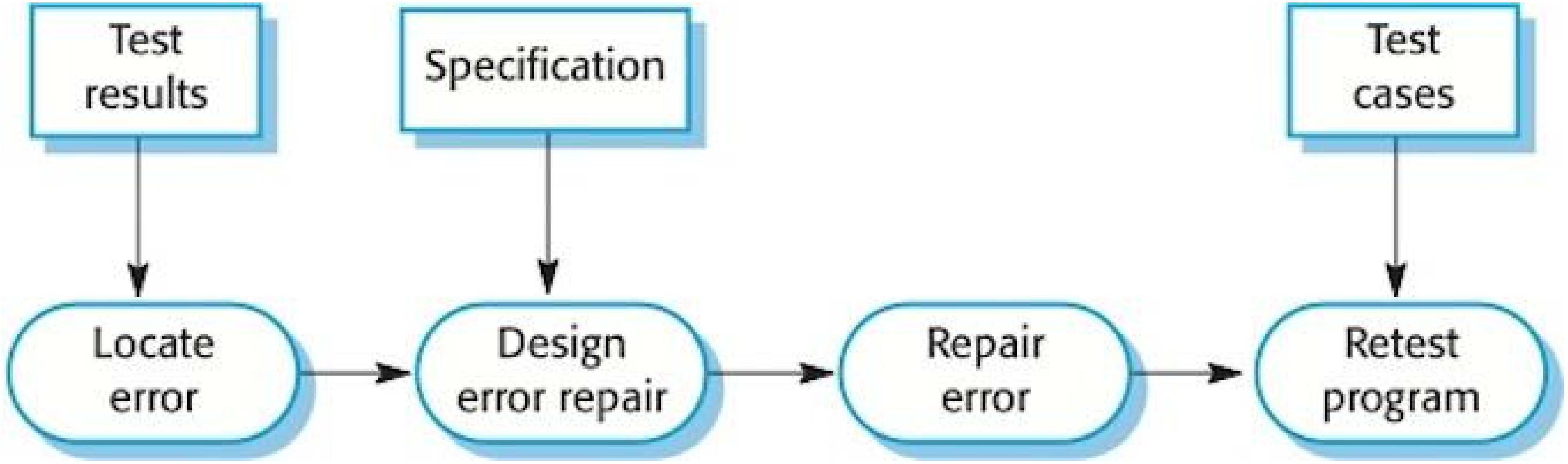
Debugging is a sanitization procedure consisting of:

- ✓ Preventing bugs in the first place, through good practices and assistive tooling.
- ✓ Detecting bugs when they first arise, through proper error handling and testing.
- ✓ Diagnosing and locating bugs, through the scientific method and interactive tooling.
- ✓ Treating bugs, through reasoned analysis and persistent refactoring.

What is debugging?

Debugging is the process of finding and resolving defects or problems within a computer program that prevent correct operation of computer software or a system.

~Wikipedia



The debugging process

Debugging Process

1. **Identifying the bug**
    - Example: A user reports that the application crashes when they click a specific button.
2. **Reproducing the issue**
    - Example: Developers try to recreate the issue by following the user's steps.
3. **Isolating the problematic code**
    - Example: By narrowing down the code, you identify the button's click event handler.
4. **Analyzing the code**
    - Example: You review the code, checking for potential issues like null references.
5. **Fixing the bug**
    - Example: You identify that a null object was accessed and correct the issue.
6. **Testing the fix**
    - Example: After the fix, the application no longer crashes, and the button works as expected.

Debugging Tools

- ✓ IDEs like Visual Studio or PyCharm provide debugging features like breakpoints and variable inspection.
- ✓ Debuggers allow you to step through code and examine variables.
- ✓ Profilers help identify performance bottlenecks in your code..

DEBUGGING TECHNIQUES

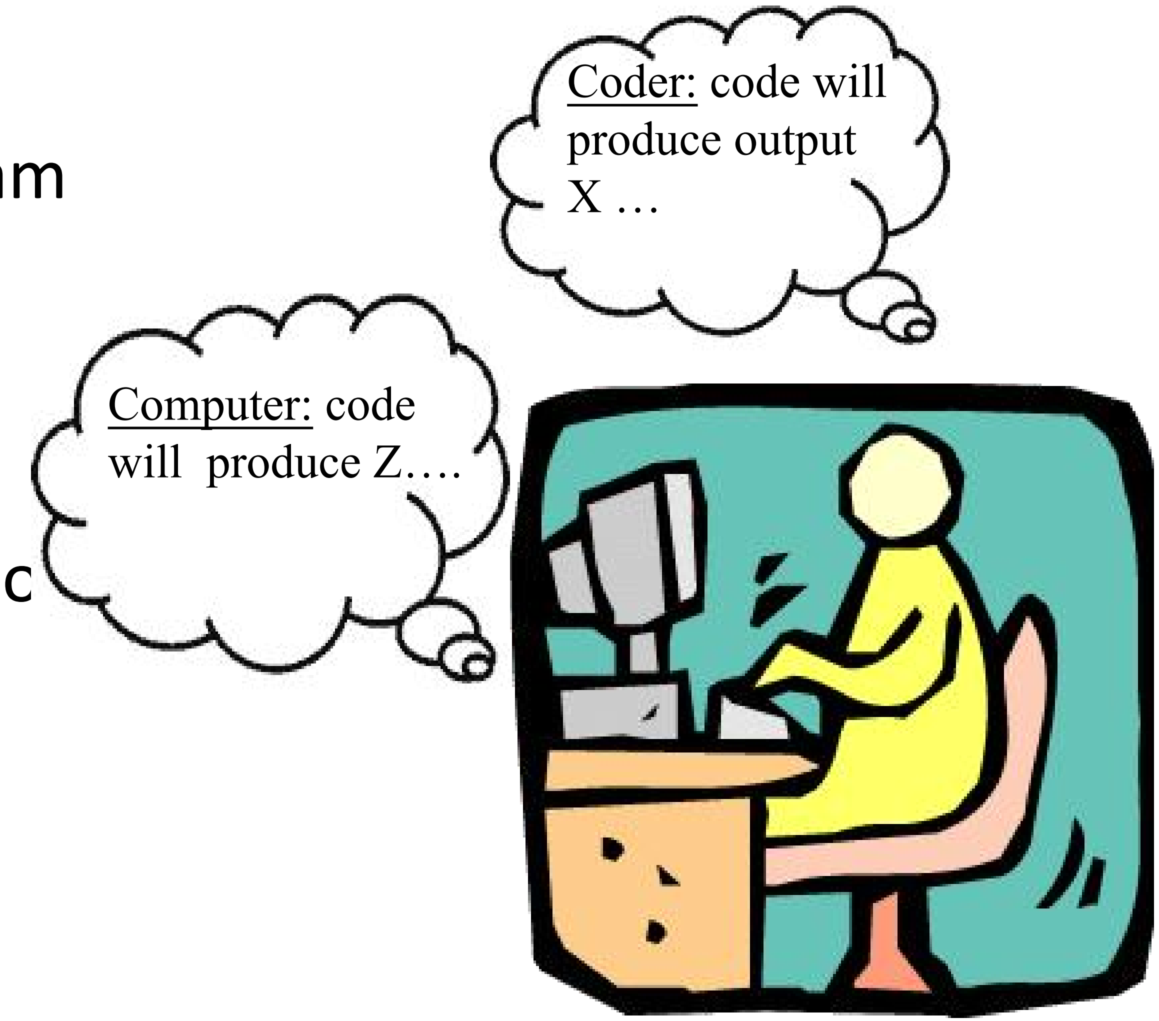
1. **Print statements**
   - Example: Adding Console.WriteLine statements to trace program flow.
2. **Breakpoints**
   - Example: Setting a breakpoint in the code to pause execution at a specific point.
3. **Logging**
   - Example: Using log files to record events, errors, and variables during program execution.
4. **Unit testing**
   - Example: Writing unit tests to verify the correctness of individual functions or methods.

Debugging Philosophy

Guiding Steps:

1) Think about why you believe the program should produce the output you expected.

2) Make assertions until you understand hc your view differs from the computer's.



Rules to Remember

- ✓ If you are lost or confused, randomly changing things around in code will not help
- ✓ You cannot fix the error if you don't know roughly where it is!
- ✓ Errors can only be caught and fixed by debugging in a systematic fashion

The unfortunate news is that sometimes it's almost impossible to figure out some problems through

DEBUGGING STRATEGIES...

Strategy #1: Debug with Purpose

Don't just change code and "hope" you'll fix the problem!

Instead, make the **bug reproducible**, then use methodical "Hypothesis Testing":

While(bug) {

1. Ask, what is the simplest input that produces the bug?
2. Identify assumptions that you made about program operation that could be false.
3. Ask yourself "How does the outcome of this test/change guide me toward finding the problem?"
4. Use pen & paper to stay organized!

}

Strategy #2: Explain it to Someone Else

- ✓ Often explaining the bug to "someone" unfamiliar with the program forces you to look at the problem in a different way.

Strategy #3: Focus on Recent Changes

If you find a NEW bug, ask:
what code did I change recently?

This favours:

- writing and testing code incrementally
- using ‘svn/git diff' to see recent changes
- regression testing (making sure new changes don't break old code).

STRATEGY #4: WHEN IN DOUBT, DUMP STATE

In complex programs, reasoning about where the bug is, can be hard, and stepping through in a debugger time-consuming.

Sometimes its easier to just “dump state” and scan through for what seems “odd”.

Example:

Dumping all packets using tcpdump.

STRATEGY #4: WHEN IN DOUBT, DUMP STATE

tools for analyzing dump states include:

GDB: The GNU Debugger is a powerful open-source debugger for analyzing crash dumps, attaching to live processes, and debugging code.

WinDbg: A Windows Debugger from Microsoft, used for debugging Windows applications and system components. It is particularly useful for analyzing Windows crash dumps.

ProcDump: A Windows tool from Sysinternals that is commonly used to capture process dumps when specific conditions are met.

Java Diagnostic Tools: For Java applications, tools like **jmap, jstack,** and profilers like **VisualVM** are often used for diagnosing issues.

Crashlytics (by Firebase): A popular crash reporting tool for mobile app development, which automatically captures crash dumps and provides analysis.

Strategy #5: Get some distance...

Sometimes, you can be TOO CLOSE to the code to see the problem.

Go for a run, take a shower, whatever relaxes you but let's your mind continue to spin in the background.

STRATEGY #6: LET OTHERS WORK FOR YOU!

Sometimes, error detecting tools make certain bugs easy to find. We just have to use them.

- runtime tools to detect memory errors

Strategy #7: Think Ahead

Bugs often represent your misunderstanding of a software interface.

Once you've fixed a bug:

1) Smile and do a little victory dance....

2) Think about if the bug you fixed might manifest itself elsewhere in your code.

3) Think about how to avoid this bug in the future

(maybe coding 36 straight hours before the deadline isn't the most efficient approach....)

FINAL WORDS

Debugging is one of the more frustrating parts of programming. It has elements of problem solving or brain teasers, coupled with the annoying recognition that you have made a mistake…

DON'T BE AFRAID TO ASK YOUR FELLOW MATES! FOR DEBUGGING HELP.

DEBUG A PROGRAM

How to Debug in eclipse/visual studio/NetBeans….?

THE RISING COST OF FIXING A BUG

What do you think fixing a bug would cost on average?

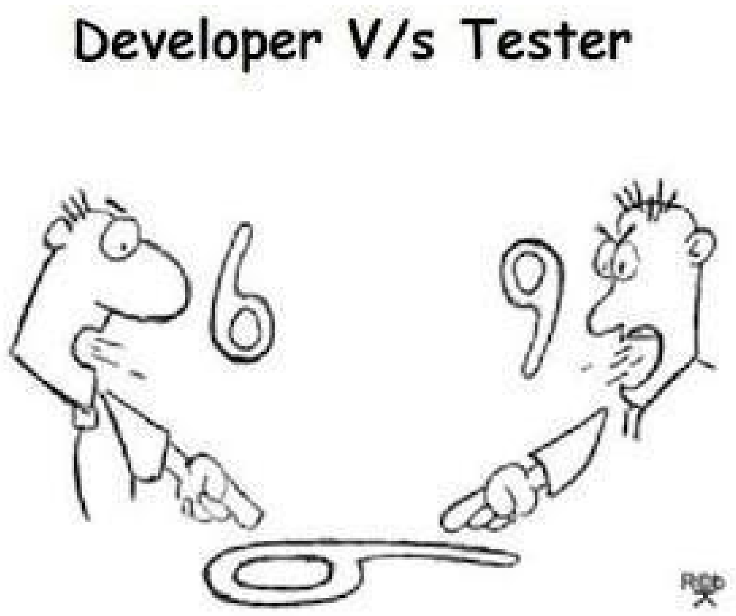
It depends on how late you find it..

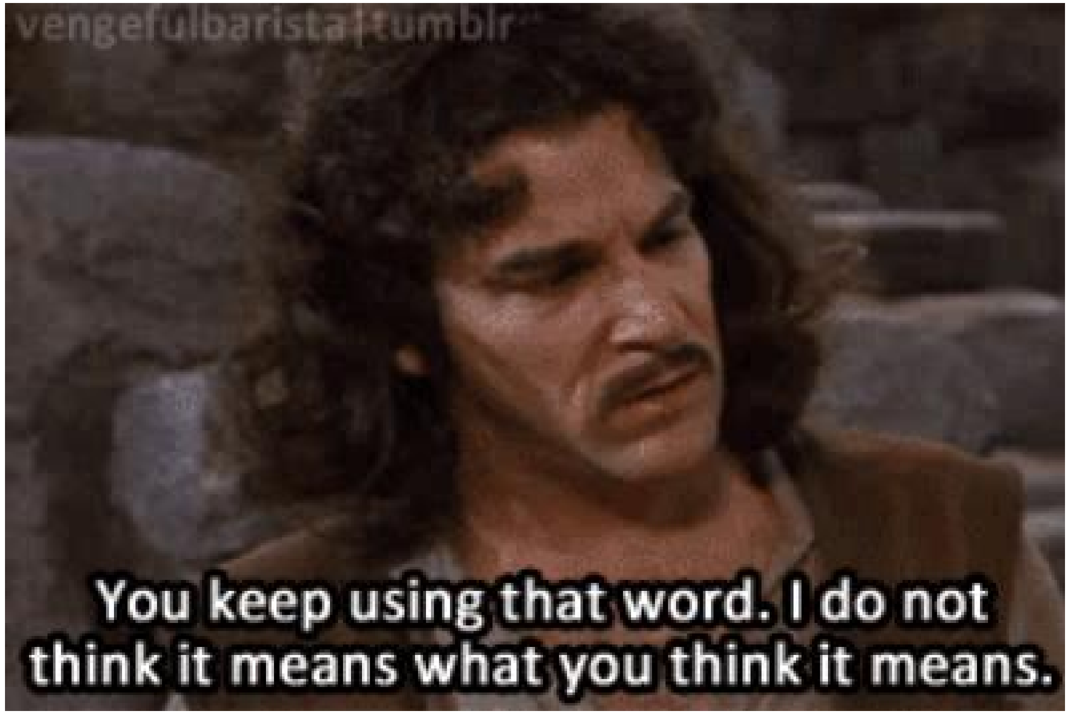
✓ The Systems Sciences Institute at IBM has reported that

- "the cost to fix an error found after product release was **four to five** times as much as one uncovered during design, and up to **100 times** more than one identified in the maintenance phase."

Considering these numbers, surely it's better to find out about such issues as soon as you possibly can!

UNIT TESTING





Testing helps increase our confidence in our code -
"If it isn't tested, assume it doesn't work"

WHAT IS UNIT TESTING

The testing that programmers do is generally called *unit testing.*

but we prefer not to use this term.

A *unit test* is a piece of code that invokes a unit of work and checks one specific end result of that unit of work. If the assumptions on the end result turn out to be wrong, the unit test has failed.

Unit testing is understood as **testing small parts of the code in isolation**.

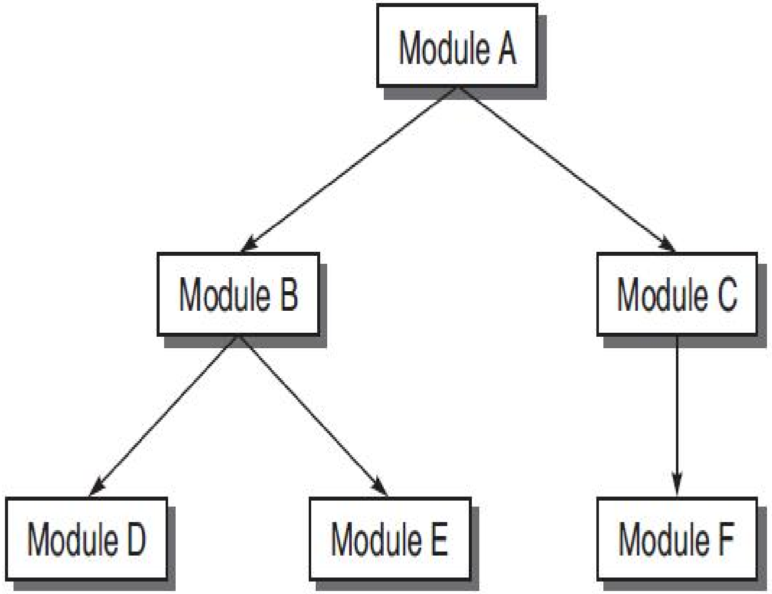
The unit under consideration might be getting some inputs from another unit or the unit is calling some other unit. It means that a unit is not independent and cannot be tested in isolation.

While testing a unit, all its interfaces must be simulated if the interfaced units are not ready at the time of testing the unit under consideration.

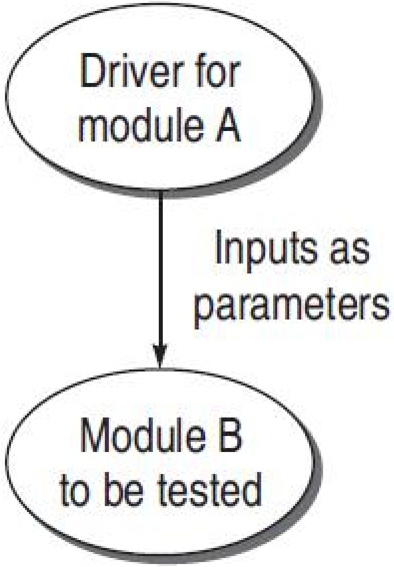
THING TO KNOW

Driver:

- Sometimes unit to be tested need input form other unit/module and that unit may not be implemented or under development
- In such a situation, we need to simulate the inputs required in the module to be tested.
- This module where the required inputs for the module under test are simulated for the purpose of module or unit testing is known as a driver module.



Suppose module A is not ready and B has to be unit tested. In this case, module B needs inputs from module A. Therefore, a **driver module** is needed which will simulate module A in the sense that it passes the required inputs to module B



THING TO KNOW

A test driver may take inputs in the following form and call the unit to be tested:

- ✓ It may hard-code the inputs as parameters of the calling unit.
- ✓ It may take the inputs from the user.
- ✓ It may read the inputs from a file.

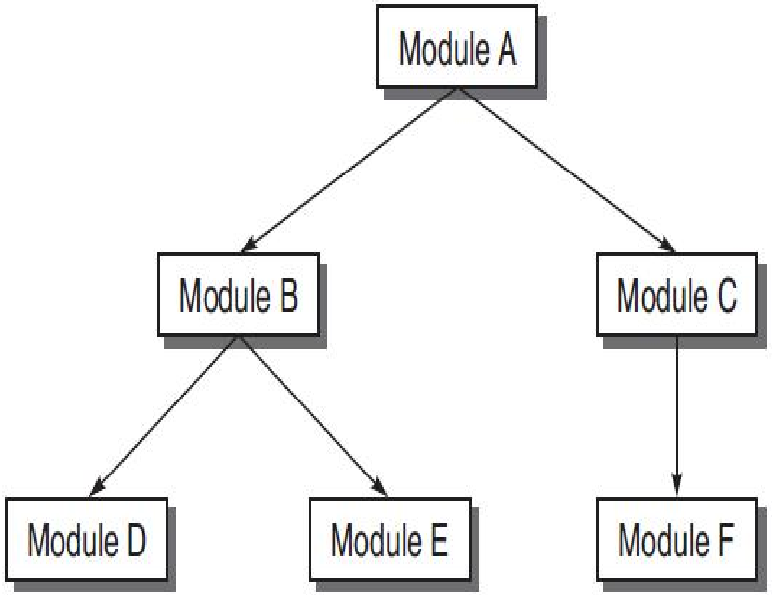
A test driver provides the following facilities to a unit to be tested:

- ✓ Initializes the environment desired for testing.
- ✓ Provides simulated inputs in the required format to the units to be tested.

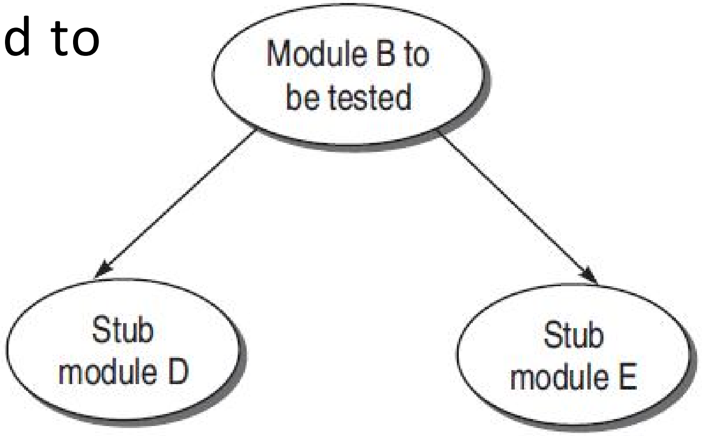
THING TO KNOW

Stubs:

- The module under testing may also call some other module which is not ready at the time of testing.
- these modules also need to be simulated for testing..
- For this dummy modules are prepared which is known as stubs.



Module B under test needs to call module D and module E. But they are not ready. So there must be some skeletal structure in their place so that they act as dummy modules in place of the actual modules. Therefore, **stubs** need to be designed for module D and module E



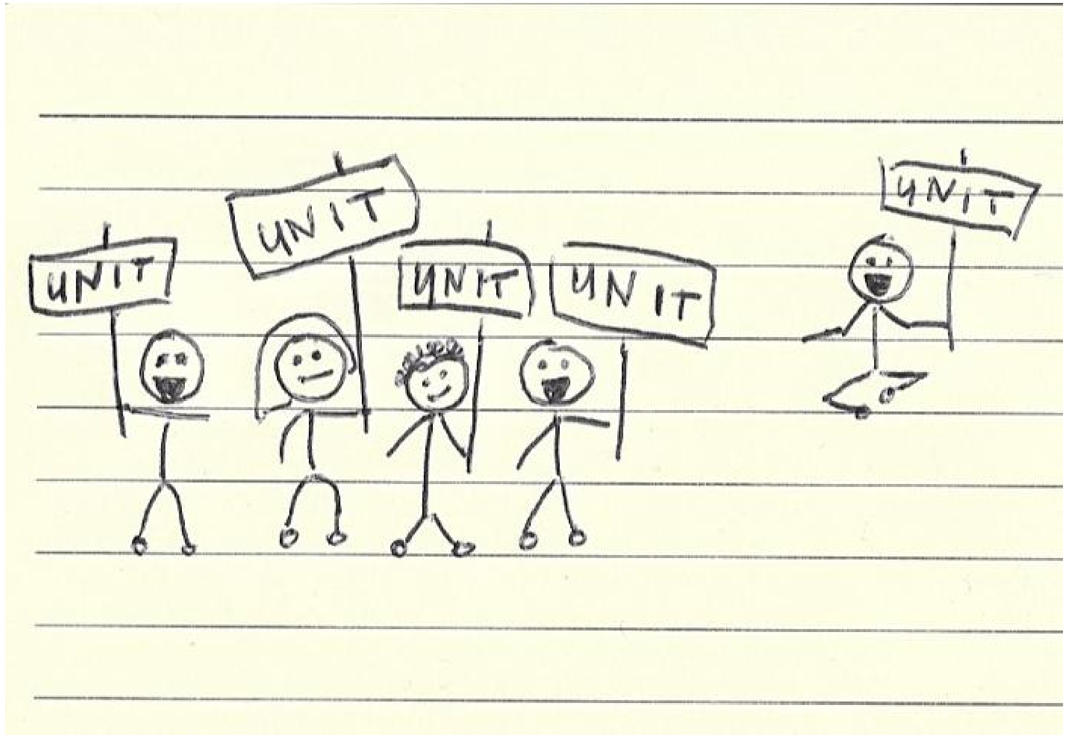
EXAMPLE

```
main()
{
	int a,b,c,sum,diff,mul;
	scanf("%d %d %d", &a, &b, &c);
	sum = calsum(a,b,c);
	diff = caldiff(a,b,c);
	mul = calmul(a,b,c);
	printf("%d %d %d", sum, diff, mul);
}
calsum(int x, int y, int z)
{
	int d;
	d = x + y + z;
	return(d);
}
```

(a) Suppose *main()* module is not ready for the testing of *calsum()* module.
Design a driver module for *main().*
(b) Modules *caldiff()* and *calmul()* are not ready when called in *main().*
Design stubs for these two modules.

How to identify a unit?

- ✓ It’s entirely up to us!
- ✓ There are no hard and fast rules for this.
- ✓ We can consider a bunch of functions as one unit or test a single function as a separate unit.



But We can use a very simple rule:

If a unit proves difficult to test, then it’s very likely that it needs to be broken down into smaller components.

PROPERTIES OF A GOOD UNIT TEST

A unit test should have the following properties:

- ✓ It should be easy to implement.
- ✓ Anyone should be able to run it at the push of a button.
- ✓ It should run quickly.
- ✓ It should be consistent in its results (it always returns the same result if you don't change anything between runs).
- ✓ It should have full control of the unit under test.
- ✓ It should be fully isolated (runs independently of other tests).
- ✓ When it fails, it should be easy to detect what was expected and determine how to pinpoint the problem.

Writing Good Tests

- Goal: to expose problems!
  - ✓ Assume role of an adversary
  - ✓ Failure == success
- Test boundary conditions
  - ✓ 0, Integer.MAX_VALUE, empty array
- Test different categories of input
  - ✓ positive, negative, and zero
- Test different categories of behavior
  - ✓ each menu option, each error message
- Test “unexpected” input
  - ✓ null pointer, last name includes a space
- Test representative “normal” input
  - ✓ random, reasonable values

EXCUSES FOR NOT TESTING

- It takes too long to run the tests
- It's not my job to test my code
- I don't really know how the code is supposed to behave so I can't test it
- But it compiles!
- I'm being paid to write code, not to write tests
- I feel guilty about putting testers and QA staff out of work
- My company won't let me run unit tests on the live system

The big Excuse

Tests are expensive to write

Yes, testing is expensive in most of industries: think about testing a home appliance, a drug or a new car…

- Test are costly
- They do **require extra effort**,
- but they improve quality, bring fast feedback, secure knowledge for newcomers…

Your First Unit Tests

To check if code is behaving as you expect, you use an ***assertion***, a simple method call that verifies that something is true.

For instance, the method ***assertTrue*** checks that the given boolean condition is true, and fails the current test if it is not.

It might be implemented like the following.

```
public void assertTrue(boolean condition) {
        if (!condition) {
                abort();
              }
        }
```

You could use this assert to check all sorts of things, including whether numbers are equal to each other:

```
int a = 2;
xx xxx xx x xxx x;
x x x xx xxx xxxx x;
assertTrue(a == 2);
xxxx xx xx xxx xx;
```

If for some reason a does not equal 2 when the `assertTrue()` is called, then the program will abort.

Your First Unit Tests

To check that two integers are equal, for instance, we could write a method that takes two integer parameters:

```
public void assertEquals(int a, int b)
{    assertTrue(a == b);
}
```

TESTING A SIMPLE METHOD

We'll start with a simple example, a single, static method designed to find the largest number in a list of numbers:

```
int Largest.largest(int[] list);
```

In other words, given an array of numbers such as [7, 8,9], this method should return 9.

That's a reasonable first test.

Think about more cases….

TESTING A SIMPLE METHOD

- How many tests did you come up with?

Your test cases should be as “What you pass in” → “What you expect”
[7, 8, 9] → 9
[8, 7, 9] → 9
[9, 7, 8] → 9

What happens if there are duplicates: [7, 9, 8, 9] → 9

What happens if there is only one number: [1] → 1

What happens with negative numbers: [-7, -9, -8, -9] → -7

To make all this more concrete, lets write a “largest” method and test it.

TESTING A SIMPLE METHOD

```
public class Largest  {
        /**
          * Return the largest element in a list.
          * @param list A list of integers
          * @return The largest number in the given list
          */
        public static int largest(int[] list) {
                int index, max=Integer.MAX_VALUE;
                for (index = 0; index < list.length-1; index++)
                        if (list[index] > max)  {
                                max = list[index];
                        }
                }
                return max;
        }
}
```

Now that we've got some ideas for tests, we'll look at writing these tests in Java, using the JUnit framework

TESTING A SIMPLE METHOD

First, let's just test the simple case of passing in a small array with a couple of unique numbers. Here's the complete source code for the test class.

```
import junit.framework.*;
public class TestLargest extends TestCase {
        public TestLargest(String name) {
                super(name);
        }
        public void testSimple() {
                assertEquals(9, Largest.largest(new int[] {7,8,9}));
        }
}
```

TESTING A SIMPLE METHOD

Having just run that code, you probably saw an error similar to the following:

```
There was 1 failure:
1)
testSimple(TestLargest)junit.framework.AssertionFailedError:
expected:<9> but was:<2147483647>
at TestLargest.testSimple(TestLargest.java:11)
```

Whoops! That is not expected. Why does it return such a huge number instead of 9?

oh, it's a small typo: `max=Integer.MAX VALUE` on line 11 should have been `max=0`. We want to initialize max so that any other number instantly becomes the next max. Let’s fix the code, recompile, and run the test again to make sure that it works.

TESTING A SIMPLE METHOD

Next we'll look at what happens when the largest number appears in different places in the list first or last, and some

```
import junit.framework.*;
public class TestLargest extends TestCase {
  public TestLargest(String name) {
    super(name);
  }
  public void testSimple() {
    assertEquals(9, Largest.largest(new int[] {7,8,9}));
  }
  public void testOrder() {
    assertEquals(9, Largest.largest(new int[] {9,8,7}));
  }
}
```

it works☺

TESTING A SIMPLE METHOD

Now try it with the 9 in the middle (just add an additional assertion the existing testOrder() method):

```
public void testOrder() {
  assertEquals(9, Largest.largest(new int[] {9,8,7}));
  assertEquals(9, Largest.largest(new int[] {7,9,8}));
}
```

it works☺

One more, just for the sake of completeness, and we can move on to more

```
public void testOrder() {
  assertEquals(9, Largest.largest(new int[] {9,8,7}));
  assertEquals(9, Largest.largest(new int[] {7,9,8}));
  assertEquals(9, Largest.largest(new int[] {7,8,9}));
}
```

```
There was 1 failure:
1) testOrder(TestLargest)junit.framework.AssertionFailedError:
   expected:<9> but was:<8>
   at TestLargest.testOrder(TestLargest.java:10)
```

Why did the test get an 8 as the largest number?

TESTING A SIMPLE METHOD

Why did the test get an 8 as the largest number?

It seems the code ignored the last entry in the list. Sure enough, another simple typo: the `for` loop is terminating too early.
Our code has:

```
for (index = 0; index < list.length-1; index++) {
```

But it should be one of :

```
for (index = 0; index <= list.length-1; index++) {
for (index = 0; index < list.length; index++) {
```

TESTING A SIMPLE METHOD

Now you can check for duplicate largest values:

```
public void testDups() {
  assertEquals(9, Largest.largest(new int[] {9,7,9,8}));
}
```

it works☺

Now the test for just a single integer:

```
public void testOne() {
    assertEquals(1, Largest.largest(new int[] {1}));
}
```

it works☺

We are almost done except the negative test

TESTING A SIMPLE METHOD

Now you can check for Negative values:

```
public void testNegative() {
  int [] negList = new int[] {-9, -8, -7};
  assertEquals(-7, Largest.largest(negList));
}
```

```
There was 1 failure:
1) testNegative(TestLargest)junit.framework.AssertionFailedError:
   expected:<-7> but was:<0>
   at TestLargest.testNegative(TestLargest.java:16)
```

How is the actual result zero?

Looks like choosing 0 to initialize max was a bad idea; what we really wanted was MIN VALUE, so as to be less than all negative numbers as well:

```
max = Integer.MIN_VALUE
```

TESTING A SIMPLE METHOD

Unfortunately, the initial specification for the method "largest" is incomplete, as it does not says what should happen if the array is empty.
We need to handle that…

Let's say that it's an error, and add some code at the top of the method that will throw a runtime-exception if the list length is zero:

```
public static int largest(int[] list) {
  int index, max=Integer.MAX_VALUE;
  if (list.length == 0) {
    throw new RuntimeException("Empty list");
  }
  ...
```

Now for the last test, we need to check that an exception is thrown when passing in an empty array.

TESTING A SIMPLE METHOD

it works☺

```
public void testEmpty() {
  try {
    Largest.largest(new int[] {});
    fail("Should have thrown an exception");
  } catch (RuntimeException e) {
    assertTrue(true);
  }
}
```

We started with a very simple method and come up with a couple of interesting tests that actually found some bugs.

UNIT TESTING BEST PRACTICES

- ✓ Unit Test cases should be independent. In case of any enhancements or change in requirements, unit test cases should not be affected.
- ✓ Test only one code at a time.
- ✓ Follow clear and consistent naming conventions for your unit tests
- ✓ In case of a change in code in any module, ensure there is a corresponding unit Test Case for the module, and the module passes the tests before changing the implementation
- ✓ Bugs identified during unit testing must be fixed before proceeding to the next phase in SDLC

Why Good Developers Write Bad Unit Tests

Utilize 15 mins to read : https://mtlynch.io/good-developers-bad-tests/